# METACHRONAL CASCADE FLAPPER

Extended Maneuver Guide — Supplement Volume

*Eight-Wing Staggered Cascade Ornithopter — Drone Prototype*

*Quadcopter maneuvers · Natural flyer behaviours · Standard ornithopter maneuvers · Bat and insect-inspired flight · Maneuvers unique to this design*

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| **Supplement scope** | This volume extends the Wave Function and Maneuver Control Guide with four new maneuver categories and a section on maneuvers unique to this architecture |
| **Chapter 10** | Quadcopter FPV maneuvers — freestyle and racing |
| **Chapter 11** | Natural flyer behaviours — birds, ornithopters, pigeoning, soaring, stooping |
| **Chapter 12** | Bat and insect-inspired flight — dragonfly, bat, hawkmoth, swift |
| **Chapter 13** | Maneuvers unique to this design — no conventional aircraft equivalent |
| **Chapter 14** | Extended quick-reference table — all new maneuvers |
| **Key principle** | Every maneuver is achieved by adjusting wave parameters in software. The 1 Hz cycle rate means entries and exits are deliberate — but any maneuver can be arrested at any point and held as a stable hover. |

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# CHAPTER 10 — QUADCOPTER FPV MANEUVERS

The following maneuvers are drawn from the FPV freestyle and racing quadcopter repertoire. They are generally achievable with the metachronal cascade architecture. The main difference from a quadcopter is that the 1 Hz cycle rate means all attitude changes are more deliberate and the aircraft can arrest any maneuver into a stable hover at any point — something a quadcopter cannot always do cleanly.

## QUADCOPTER — BASIC FREESTYLE

*Power loops, flips, rolls, yaw spins*

QUADCOPTER FREESTYLE

### POWER LOOP

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| **Mechanisms used** | Full-throttle vertical loop — aircraft pitches forward into a loop while maintaining power throughout. In quadcopter terms: punch throttle, pitch forward continuously through 360 degrees. On this aircraft: sustained nose-down pitch input throughout the loop while maintaining climb power. The metachronal wave maintains lift even in inverted phase. |
| **Wave parameters** | Throttle: maximum throughout. Pitch: continuous nose-down input. Phase: counter-phase for smooth lift. Both-stroke AoA activated at inverted. Recovery: return to forward flight AoA at bottom. |
| **Expected result** | Vertical 360-degree loop with power maintained. Diameter 10–20 m. Altitude neutral if entry and exit speeds matched. Slower and more graceful than a quadcopter power loop — the wing-based lift gives a smoother arc. |

QUADCOPTER FREESTYLE

### FRONT FLIP

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| **Mechanisms used** | Rapid 360-degree pitch rotation. In quadcopter terms: momentary full pitch forward input through a complete rotation. On this aircraft the 1 Hz cycle rate means a full flip takes 3–5 seconds — it is a deliberate maneuver, not an instantaneous snap. The aircraft passes through inverted and recovers. Both-stroke AoA essential to maintain altitude during inverted phase. |
| **Wave parameters** | Pitch: sustained forward input. Throttle: increased during recovery phase. Both-stroke AoA: activated as bank exceeds 90 degrees. Recovery: pitch reverses as nose returns to horizon. |
| **Expected result** | 360-degree pitch rotation in 3–5 seconds. Altitude loss approximately 2–4 m if not actively managed. Smooth exit to level hover. Characteristic slow-motion quality compared to quadcopter flip. |

QUADCOPTER FREESTYLE

## BACK FLIP

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| **Mechanisms used** | Identical to front flip with pitch direction reversed. Nose pitches backward, aircraft rotates tail-over-nose in the reverse direction. Requires same both-stroke AoA activation during inverted phase. Slightly more disorienting visually — camera initially points skyward. |
| **Wave parameters** | Pitch: sustained nose-up input. Throttle: maintained. Both-stroke AoA: activated through inverted. Recovery: pitch reverses as nose returns to horizon from below. |
| **Expected result** | 360-degree backward pitch rotation. Recovery to hover. Combined with yaw — 'back flip into spin' — is a natural combo sequence. |

QUADCOPTER FREESTYLE

## YAW SPIN

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| **Mechanisms used** | Continuous rotation about the vertical axis while maintaining position and altitude. In quadcopter terms: full yaw input held continuously. On this aircraft: full differential wave phase input maintained. This is one of the cleanest maneuvers for this architecture — the yaw mechanism is very effective and pure. |
| **Wave parameters** | Yaw: maximum differential phase, left side leads right by 90–180 degrees. Throttle: hover setting maintained. Roll and pitch: neutral. Position: held by GPS if available. |
| **Expected result** | Smooth continuous rotation. Rate 45–80 degrees/second. Full 360 degrees in 5–8 seconds. No altitude loss. No translation. Can be combined with throttle changes for dramatic visual effect. |

QUADCOPTER FREESTYLE

## INVERTED HOVER

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| **Mechanisms used** | Sustained flight upside-down at a fixed position. In quadcopter terms: hold inverted with throttle reversed. On this aircraft: both-stroke AoA activated with upper surface of wing now becoming the lifting surface. The symmetric airfoil variant (NACA 0015 + VG + coverts) is required — S1223 has limited inverted lift capability. |
| **Wave parameters** | Bank: 180 degrees (fully inverted). Both-stroke AoA: both strokes at 18–22 degrees (both generate lift on what is now the lower surface — former upper surface). Throttle: increased 25–30% versus upright hover. Pitch and roll corrections: reversed in pilot reference frame. |
| **Expected result** | Sustained inverted hover. Duration limited by power — approximately 40–60% less efficient than upright hover. Requires symmetric airfoil. Unique characteristic: smoother than quadcopter inverted hover due to continuous lift wave. |

QUADCOPTER FREESTYLE

## MATTY FLIP

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| **Mechanisms used** | Fly backward over an obstacle, invert, complete a backward loop and exit going forward. Entry in forward flight toward an obstacle. Pull nose up as the obstacle is passed, continue pulling through inverted, complete the arc and exit in the forward direction. The metachronal wave handles the inversion automatically — only pitch input is required from the pilot. |
| **Wave parameters** | Entry: forward cruise. Pitch: maximum nose-up as obstacle is cleared. Over the top: both-stroke AoA, throttle maintained. Pull through: continued nose-up through inverted and back to upright. Exit: return to forward cruise. |
| **Expected result** | Backward arc over an obstacle exiting in the original direction but having reversed and returned. Camera arc is dramatic. Altitude at apex equals obstacle height plus pull-up radius. |

QUADCOPTER FREESTYLE

## SPLIT-S INTO HOVER

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| **Mechanisms used** | Half roll to inverted, pull through to upright — but instead of exiting in forward flight, reduce throttle and transition to hover at the bottom of the arc. Unique to this architecture — a quadcopter cannot transition cleanly from the bottom of a Split-S pull-out into a precise hover because the attitude change happens too quickly. The 1 Hz cycle rate here becomes an advantage — the pull-out is slow enough to bleed speed into hover. |
| **Wave parameters** | Phase 1 — half roll: full roll input to 180 degrees. Phase 2 — pull-through: maximum pitch input. Phase 3 — speed decay: reduce throttle as airspeed falls. Phase 4 — hover transition: wave returns to counter-phase, throttle to hover setting. |
| **Expected result** | Clean arrest from forward flight through the Split-S maneuver directly into a stable hover at the bottom. No oscillation. Highly impressive visually — appears to defy momentum. |

# QUADCOPTER — ADVANCED FPV

*Combo sequences and precision moves*

QUADCOPTER ADVANCED

## PUNCH OUT

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| **Mechanisms used** | Full-throttle vertical climb — maximum power burst straight up. In quadcopter terms: punch throttle from hover to maximum. On this aircraft: all wings switch to in-phase mode (0/0/0/0 degrees), maximum amplitude, maximum AoA on both strokes simultaneously. The cascade generates maximum instantaneous lift for a vertical surge. |
| **Wave parameters** | Phase: all wings in-phase (0 degrees all). Amplitude: maximum (70 degrees). AoA: both-stroke mode, 42 degrees power, 18 degrees recovery. Duration: brief — 3–5 seconds maximum before overheating servos. |
| **Expected result** | Aggressive vertical climb rate: 3.5–5.5 m/s burst. High body oscillation during this mode. Aircraft surges upward in a way unlike normal climb — the in-phase pulse is viscerally powerful. |

QUADCOPTER ADVANCED

## INVERTED YAW SPIN

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| **Mechanisms used** | Continuous yaw rotation while inverted. The combination of inverted hover (both-stroke AoA activated) with full differential wave phase for yaw. Requires symmetric airfoil. The most mechanically demanding sustained maneuver in the repertoire — high power draw in inverted mode plus the continuous phase differential for yaw. |
| **Wave parameters** | Bank: 180 degrees inverted. Both-stroke AoA: activated (18–22 degrees). Yaw: full differential phase maintained. Throttle: hover-plus (25–30% above normal hover). |
| **Expected result** | Inverted rotation. Rate similar to upright yaw spin (45–80 deg/s). Visually spectacular. Duration limited by power — approximately 8–12 minutes maximum endurance in this combined mode. |

QUADCOPTER ADVANCED

## GAP FLYING — PRECISION TRANSIT

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| **Mechanisms used** | Flying through a narrow gap — between branches, through a window, under a bridge. The sweep-to-stow capability is unique here: outer wings (W3, W4) can be partially swept back to reduce effective wingspan for the transit, then re-deployed on exit. No quadcopter can reduce its footprint mid-flight. The gap can be narrower than the full deployed wingspan. |
| **Wave parameters** | Approach: wings at full span. Gap transit: W3 and W4 swept to 60–80 degrees aft — wingspan reduces by 35%. Throttle: maintained at cruise. Exit: W3 and W4 re-deployed to full span. |
| **Expected result** | Aircraft can transit gaps 35% narrower than full deployed wingspan. This is a unique capability. Transition in and out of reduced span is smooth — the wave generator automatically recalculates phase for the partially stowed configuration. |

QUADCOPTER ADVANCED

## ORBIT — INVERTED ORBIT

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| **Mechanisms used** | Circle a point of interest while maintaining it centred in the FPV view. Achieved by combining forward flight (stroke plane tilt toward centre) with continuous yaw tracking the orbit direction. Inverted orbit: the same maneuver performed upside-down. The aircraft orbits a point while inverted, camera pointing down at it. |
| **Wave parameters** | Forward: stroke plane tilted toward orbit centre at orbit radius speed. Yaw: continuous, rate matched to orbit circumference and speed. Inverted orbit: add both-stroke AoA to above. Altitude: maintained by throttle feedback. |
| **Expected result** | Smooth orbit. Radius adjustable from 3 m to 30 m. Inverted orbit is highly unusual — few aircraft can sustain it cleanly. Continuous wave lift makes this significantly smoother than equivalent quadcopter orbit. |

# CHAPTER 11 — NATURAL FLYER BEHAVIOURS — BIRDS AND ORNITHOPTERS

The following maneuvers are derived from observed behaviours of natural flyers and standard ornithopter demonstrations. Each has a direct mapping to metachronal wave parameter adjustments.

## BIRD-INSPIRED MANEUVERS

*Pigeon, soaring birds, falcon, swift*

BIRD-INSPIRED

### PIGEONING — SLOW MANOEUVRING TURN

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| **Mechanisms used** | Pigeons at low speed use body rotation to redirect aerodynamic force rather than using conventional banking. **Research shows pigeons steer like helicopters at low speeds, generating lift on both upstroke and downstroke during turns.** This architecture replicates this exactly: at low speed, yaw the body toward the turn while maintaining symmetric lift from both sides using both-stroke AoA. The turn is achieved by body reorientation, not banking. |
| **Wave parameters** | Speed: very low (hover-to-3 km/h). Yaw: differential phase to orient body toward turn. Roll: minimal — body turns, wings stay near level. Both-stroke AoA: activated to maintain lift during yaw. Throttle: hover setting. |
| **Expected result** | 90-degree heading change at near-zero speed. Aircraft pivots on the spot while drifting slowly in the new direction. Very precise — useful for obstacle avoidance at low speed. Exactly replicates the kinematic strategy of Columba livia in slow flight. |

BIRD-INSPIRED

### THERMAL SOARING

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| **Mechanisms used** | Exploit a thermal (rising column of warm air) to gain altitude with wings fixed or near-fixed, using zero or minimal flapping power. Wings locked at optimal glide AoA (12 degrees), aircraft circles within the thermal using gentle banked turns. Flapping resumes only if the aircraft drifts to the edge of the thermal and needs to return to centre. This is the albatross and raptor strategy for near-zero-energy cruising. |
| **Wave parameters** | Wing stroke: stopped or near-stopped (glide mode). Wing AoA: fixed at 12 degrees (optimal glide). Orbit: gentle banked turn within thermal column, radius 15–40 m. Flap bursts: brief, as needed to maintain orbit position. |
| **Expected result** | Altitude gain at near-zero power draw — the thermal does the work. In a 1 m/s thermal, altitude gain rate equals thermal rate minus sink rate. Estimated sink rate in glide: 0.3–0.8 m/s. In a strong thermal (2–3 m/s) the aircraft climbs at 1.2–2.2 m/s with wings nearly stationary — extraordinary efficiency. |

BIRD-INSPIRED

## SLOPE SOARING

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| **Mechanisms used** | Exploit ridge lift — air deflected upward by a hillside, cliff, or building. Wings fixed in glide mode, aircraft flies parallel to the ridge in the rising air band. Banking away from the ridge at each end to reverse direction. No power required as long as the lift band exceeds the aircraft sink rate. This is standard ornithopter soaring behaviour. |
| **Wave parameters** | Wings: glide mode, fixed AoA. Track: parallel to ridge, in rising air band. Turns: gentle banked reversals at ridge ends. Power: zero in lift band; brief flap bursts in sink zones. |
| **Expected result** | Indefinite soaring along a ridge with minimal power. The large wing area of the 8-wing cascade gives a very low wing loading, meaning the aircraft can soar in weaker ridge lift than conventional aircraft. Comparable to a high-performance sailplane in soaring efficiency. |

BIRD-INSPIRED

## FALCON STOOP — HIGH-SPEED DIVE

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| **Mechanisms used** | Replicates the peregrine falcon's hunting dive. **The falcon adopts a teardrop shape — wings tucked — to minimise drag, then deploys wings in M-shape for the pull-out, generating vortex lift for pitch and roll control.** On this aircraft: all eight wings swept to maximum aft position (stow), fuselage becomes a streamlined projectile. At the target altitude, all wings deploy simultaneously to full forward sweep — the sudden increase in lift area generates a powerful pull-out force. |
| **Wave parameters** | Stoop entry: climb to altitude. All wings: swept to full aft (stow position). Dive: gravity-accelerated, wings stowed, minimum drag. Pull-out: all 8 wings simultaneously deployed to full forward sweep. AoA: maximum on all wings simultaneously — peak vortex lift. Recovery: transition to forward cruise or hover. |
| **Expected result** | Maximum speed achieved in stoop: estimated 45–65 km/h (limited by structural loads and servo authority). Pull-out generates significant g-loading — structural margin must be verified before executing at full dive speed. The simultaneous 8-wing deployment is a unique and dramatic visual event. |

BIRD-INSPIRED

## SWIFT POST-STALL ARREST

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| **Mechanisms used** | Rapid deceleration from high-speed forward flight using dynamic stall as an airbrake — directly observed in common swifts. **Braking possible by rapid body pitch and dynamic stall at relatively high air speed, with maximum deceleration of 31.4 m/s squared demonstrated in ornithopters.** All wings simultaneously pitched to maximum drag AoA on both strokes, stroke plane tilted aft. Wings act as parachutes. |
| **Wave parameters** | All wings: both strokes at 40–45 degrees AoA (maximum drag, both strokes). Stroke plane: tilted 35 degrees aft. Phase: counter-phase for smoothness. Duration: 1–3 seconds to arrest from cruise speed. |

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| **Expected result** | Maximum deceleration approximately 6–8 m/s squared. Speed arrested from 30 km/h to hover in approximately 1.5–3 m. Transition to stable hover at the end of the arrest. Highly useful for precision landing approaches and obstacle avoidance. |

BIRD-INSPIRED

## UNDULATING FLIGHT — BOUNDING

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| **Mechanisms used** | Many birds use an undulating flight path — flapping bursts followed by brief wing-closed glide phases, creating a sinusoidal altitude profile. This saves energy because the flap phase converts chemical energy to both kinetic energy and altitude, and the glide phase converts altitude back to forward speed. The net energy cost is lower than continuous flapping at the same average forward speed. |
| **Wave parameters** | Flap phase (0.8–1.2 seconds): normal metachronal wave, full amplitude. Glide phase (0.5–1.5 seconds): wings swept to aft cruise position, AoA fixed at 8–10 degrees, stroke stopped. Cycle: alternating flap and glide phases at pilot or autopilot command. |
| **Expected result** | Energy saving estimated 15–25% versus continuous flapping at same speed. Altitude oscillation of 0.5–2 m per cycle. Average forward speed maintained. Natural-looking flight profile — most passerine birds use this strategy. |

BIRD-INSPIRED

## PERCH LANDING — DYNAMIC STALL FLARE

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| **Mechanisms used** | Landing on a narrow perch with precision. Bird approach: decelerate to near-zero speed using a flare (rapid nose-up that creates high-drag stall), then contact the perch at near-zero forward speed. On this aircraft: combine the swift arrest maneuver with a precise hover translation to position directly over the perch, then settle vertically. If equipped with a gripper or hook, landing upright on a perch is achievable. |

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| **Wave parameters** | Approach: forward flight at reduced speed toward perch. Flare: maximum nose-up, both-stroke drag AoA to arrest forward speed. Hover: transition to stable hover directly above perch. Settle: reduce throttle, descend vertically to contact. |
| **Expected result** | Precision landing on a surface the width of the fuselage. Forward speed at contact: near zero. Vertical speed at contact: 0.2–0.5 m/s. This capability is unique among drone-class vehicles — no conventional fixed-wing aircraft can perform a perch landing. |

# CHAPTER 12 — BAT AND INSECT-INSPIRED MANEUVERS

Bats and insects demonstrate flight capabilities that go beyond the bird repertoire. Many are directly emulable on this architecture due to the independent per-wing 3-DoF control. Several have never been demonstrated on any drone platform.

## DRAGONFLY-INSPIRED

*From the world's most maneuverable insect*

DRAGONFLY-INSPIRED

### 180-DEGREE TURN IN THREE WINGBEATS

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| **Mechanisms used** | **Dragonflies can turn 180 degrees in three wingbeats.** This requires an extremely rapid yaw reversal combined with a brief in-phase power burst. On this aircraft: maximum differential wave phase (full yaw authority), simultaneously with in-phase power burst for maximum torque authority. Three 1 Hz cycles = 3 seconds for the full reversal — very fast for a vehicle of this scale. |
| **Wave parameters** | Yaw: maximum differential phase throughout. Simultaneously: all wings to in-phase for peak torque authority. Duration: 3 wing cycles (3 seconds at 1 Hz). Exit: wave returns to metachronal, forward flight AoA. |
| **Expected result** | 180-degree heading reversal in 3 seconds. Yaw rate during maneuver: approximately 60 degrees/second. Some altitude loss — in-phase burst compensates partially. Fastest possible heading reversal for this aircraft. |

DRAGONFLY-INSPIRED

### FIGURE-8 HOVER OSCILLATION

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| **Mechanisms used** | Insects in sustained hover often oscillate their position slightly in a figure-8 or oval pattern rather than holding perfectly still. **Figure-8 wing motion is more efficient at low Reynolds numbers than simple sinusoidal motion.** The deviation axis (theta) on each wing already introduces a figure-8 tip path. This maneuver deliberately amplifies that oscillation to increase lift efficiency in hover. |
| **Wave parameters** | Deviation amplitude: increased from 12 degrees to 20–25 degrees. Deviation frequency: remains at 2x flap frequency (2 Hz). Phase of deviation: tuned for peak efficiency (initial phase 60–80 degrees). All other parameters: standard hover setting. |
| **Expected result** | Hover efficiency improvement estimated 5–12% versus standard hover. Slight body oscillation is the tradeoff — acceptable for most applications. This mode is particularly effective in low-speed ground-effect hovering. |

DRAGONFLY-INSPIRED

## SIDEWAYS INTERCEPT DASH

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| **Mechanisms used** | Dragonflies intercept prey with 95% success rate using a predictive intercept course — not chasing, but flying to where the prey will be. This requires instantaneous lateral translation toward the intercept point. On this aircraft: maximum lateral crab (asymmetric stroke plane tilt) combined with forward flight to generate a diagonal velocity vector aimed at the intercept point. Adjustable in real time. |
| **Wave parameters** | Forward component: stroke plane forward tilt, proportional to intercept bearing. Lateral component: asymmetric stroke plane tilt, proportional to lateral range. Vector sum: adjusted every 2–3 cycles to track moving target. Speed: maximum cruise (38–52 km/h). |
| **Expected result** | Diagonal pursuit vector toward intercept point. Heading can remain fixed while velocity vector is oblique — crabbing toward the target while pointing forward. Useful for tracking a moving subject in FPV cinematography. |

# BAT-INSPIRED

*Inertial reorientation and inverted landing*

BAT-INSPIRED

## INERTIAL FLIP — ZERO-MOMENTUM TURN

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| **Mechanisms used** | **Bats perform zero-angular-momentum turns by differential modulation of wing inertia — retracting one wing while extending the other to rotate without aerodynamic force.** On this aircraft: one side's wings swept rapidly to full aft (stow) while the other side stays fully deployed. The asymmetric mass distribution generates an inertial torque that rotates the aircraft. This works even without aerodynamic force — it uses physics, not airflow. |
| **Wave parameters** | Left wings: rapidly swept to full aft (stow) in 0.2 seconds. Right wings: maintained at full forward sweep. Stroke: reduced or stopped during the flip. Recovery: left wings re-deploy as desired orientation is reached. |
| **Expected result** | Rapid body rotation using inertial dynamics rather than aerodynamics. Works at very low speed where aerodynamic control authority is reduced. The roll rate from inertial reorientation can exceed aerodynamic roll rate at speeds below 5 km/h. Unique — no conventional drone exploits this. |

BAT-INSPIRED

## INVERTED LANDING — PERCH UPSIDE-DOWN

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| **Mechanisms used** | Bats land upside-down on ceilings using inertial reorientation. **During landing the bat retracts wings on upstroke, extends on downstroke, then pulls one wing at the flip moment — all in 0.5 seconds.** On this aircraft: approach ceiling inverted (both-stroke AoA mode), then retract one side's wings in sequence to perform the flip to inverted, reach gripper to surface. Requires a gripper or hook attachment. |
| **Wave parameters** | Approach: inverted hover beneath ceiling target. Flip: right-side wings swept aft rapidly (inertial flip). Contact: reach gripper to surface as inversion completes. Locked: wave stops, aircraft hangs inverted from attachment. |

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| **Expected result** | Inverted ceiling attachment. No other current drone achieves this. Requires gripper hardware addition (40–80 g). Once attached, all wings stow and power reduces to zero — the aircraft hangs passively like a bat. Battery charging possible in this configuration if the attachment point is wired. |

BAT-INSPIRED

## SHARP BANKING TURN — BAT STYLE

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| **Mechanisms used** | Bats use asymmetric wing inertia for sharp banking turns through cluttered environments. The inside wing is retracted (swept aft) during the turn to reduce its moment of inertia and allow faster rotation, while the outside wing remains extended for lift. On this aircraft this maps directly to sweep asymmetry in turns. |
| **Wave parameters** | Inside wing (left in left turn): W3 and W4 swept 60 degrees aft during turn — reduced effective area and inertia. Outside wing: full forward sweep maintained for maximum lift. Yaw: differential phase adds torque to the turn. Bank: amplitude differential provides roll moment. |
| **Expected result** | Tighter turn radius than conventional banked turn — estimated 20–30% radius reduction versus pure aerodynamic turn. The combination of inertial asymmetry plus aerodynamic differential provides the fastest possible turn for this aircraft. Useful for obstacle avoidance in cluttered environments. |

BAT-INSPIRED

## HAWKMOTH HOVER-FEED OSCILLATION

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| **Mechanisms used** | Hawkmoths hover in front of flowers while feeding with a long proboscis, matching the flower's position as it sways in wind. They use extremely precise low-frequency position corrections — essentially a biological position-hold with active disturbance rejection. The wave generator implements this as a feed-forward correction based on visual or sensor input about target position. |

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| **Wave parameters** | Base mode: standard counter-phase hover. Correction loop: lateral and fore-aft stroke plane tilt, updated each wing cycle based on target position offset. Update rate: once per flap cycle (1 Hz corrections). Amplitude of correction: proportional to position error. |
| **Expected result** | Precision station-keeping relative to a moving or oscillating target. Position error maintained within 0.1–0.3 m of target. The 1 Hz correction rate matches well to the wind-induced sway frequency of most natural targets. Directly applicable to precision inspection or delivery applications. |

## INSECT ESCAPE AND EVASION

*Moth evasion, escape jink, chaos flight*

INSECT-INSPIRED

### ERRATIC MOTH EVASION PATTERN

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| **Mechanisms used** | Moths use an unpredictable stochastic flight path when detecting bat ultrasound — the erratic path is specifically designed to be impossible to predict. On this aircraft: stochastic perturbations are added to the phase offsets and amplitude of each wing at 3–5 Hz (every few stroke cycles), creating an unpredictable flight path that is impossible to intercept by a simple pursuit algorithm. This is a stealth/evasion mode. |
| **Wave parameters** | Phase: random perturbation of plus or minus 20 degrees added every 2–3 cycles to each wing independently. Amplitude: random variation plus or minus 15% each cycle. Net position: biased toward a general heading but with unpredictable lateral and vertical deviations of 0.3–1.5 m per perturbation. |
| **Expected result** | Unpredictable flight path that defeats simple intercept algorithms. Useful for counter-interception operations or training exercises. The biological inspiration is directly applicable — this pattern evolved specifically to defeat bat echolocation targeting. Extends to defeating radar tracking algorithms. |

INSECT-INSPIRED

## ESCAPE JINK — SUDDEN DIRECTION REVERSAL

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| **Mechanisms used** | Many insects perform a sudden near-instantaneous direction reversal when threatened — a single wingbeat in the reverse direction combined with a body roll. On this aircraft: trigger a single cycle of the reverse wave simultaneously with a full roll input, then return to forward flight in any chosen direction. The maneuver is completed in 2–3 seconds but produces a velocity reversal of several m/s. |
| **Wave parameters** | Single cycle: wave direction reversed (270/180/90/0). Simultaneously: maximum roll input to body-roll the aircraft. Amplitude: maximum for maximum impulse. Exit: wave returns to chosen exit direction heading. |
| **Expected result** | Rapid direction change — vehicle velocity reversed in 2–3 seconds. Exit heading can be any direction — the body roll during the reversal reorients the aircraft so exit heading is freely choosable. Effective emergency obstacle avoidance maneuver. |

# CHAPTER 13 — MANEUVERS UNIQUE TO THIS DESIGN

The following maneuvers have no equivalent in any conventional fixed-wing, rotary-wing, or existing ornithopter design. They emerge specifically from the combination of eight independently controlled 3-DoF wings, staggered cascade geometry, metachronal wave control, and the sweep-equals-stow architecture.

## UNIQUE TO THIS ARCHITECTURE

*No conventional aircraft equivalent*

UNIQUE TO THIS DESIGN

### PROGRESSIVE CASCADE STOW — ACCORDION FLIGHT

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| **Mechanisms used** | Wings are progressively stowed from rear to front during forward flight, then re-deployed front to rear — the aircraft 'breathes' like an accordion. As each wing stows, the remaining wings increase amplitude to compensate. The effective wing count cycles continuously: 8 → 6 → 4 → 6 → 8. This is not possible on any other aircraft and has no natural equivalent. It allows real-time optimisation of the wing count for current flight conditions. |
| **Wave parameters** | Stow sequence: W4 stows, W1-W3 increase amplitude by 15%. Then W3 stows, W1-W2 increase amplitude by 25%. Then W3 re-deploys, then W4 re-deploys. Cycle rate: one full accordion cycle every 8–12 seconds. |
| **Expected result** | Continuously varying aerodynamic configuration in flight. Allows real-time testing of optimal wing count for efficiency. Visually striking — the aircraft visibly changes shape in flight. No analogue in biology or engineering. Practical application: automated efficiency optimisation. |

UNIQUE TO THIS DESIGN

### WAVE REVERSAL IN HOVER — STANDING WAVE

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| **Mechanisms used** | Rather than a traveling wave (front-to-back or back-to-front), the controller switches to a standing wave — alternating pairs of wings beat in opposition (W1+W3 versus W2+W4). This creates a resonant condition where the lift pulses from the two in-phase pairs partially cancel each other — giving the smoothest possible hover with the lowest body oscillation of any mode. It is a fundamentally different wave topology from any described mode. |
| **Wave parameters** | Phase: 0/180/0/180 degrees (W1 and W3 in phase, W2 and W4 in phase but opposite to W1/W3). Left-right symmetric. Amplitude: slightly reduced (50 degrees) — smoothness over power. AoA: 32 degrees power, 5 degrees recovery. |
| **Expected result** | Minimum body oscillation hover mode — lower than counter-phase mode. Lift is generated by two pairs acting as virtual coupled oscillators. Power consumption slightly higher than counter-phase. Best mode for precision hovering in calm conditions, sensor operations, or FPV cinematography. |

UNIQUE TO THIS DESIGN

## ASYMMETRIC CASCADE — DIFFERENTIAL EFFICIENCY FLIGHT

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| **Mechanisms used** | The left and right sides of the cascade use different wave modes simultaneously. Left side: metachronal wave for efficient forward flight. Right side: counter-phase for smooth lift with minimal forward thrust. The asymmetry between sides creates a curved flight path — a continuous gentle left turn — while both sides are operating at their respective optimal efficiency points. The turn radius is set by the degree of asymmetry between the two wave modes. |
| **Wave parameters** | Left side: forward metachronal wave (0/90/180/270 degrees). Right side: counter-phase standing wave (0/180/0/180 degrees). Amplitude: equal both sides. AoA: equal both sides. |
| **Expected result** | Continuous curved flight path using different optimisation criteria on left versus right. No conventional aircraft equivalent. Useful for sustained orbit of a fixed point at maximum efficiency — the efficient side provides cruise thrust, the smooth side provides stable lift for the turn. |

UNIQUE TO THIS DESIGN

## SEQUENTIAL WING DEPLOYMENT — BURST LAUNCH

| | |
|---|---|
| **Mechanisms used** | From fully stowed (all wings at aft maximum), wings deploy sequentially from front to back — W1 deploys and immediately begins a power stroke, W2 deploys 250 ms later mid-stroke, W3 deploys 500 ms later, W4 750 ms later. Each wing adds its full power stroke immediately upon deployment. The result is a cascading burst of lift — each wing contributing a separate lift pulse in sequence — that produces a very high instantaneous lift for takeoff from a confined space. |
| **Wave parameters** | T=0: W1 deploys, begins power stroke immediately. T=250ms: W2 deploys, begins power stroke. T=500ms: W3 deploys, begins power stroke. T=750ms: W4 deploys, begins power stroke. All wings: maximum amplitude, in-phase at deployment, then metachronal wave established over next 2 cycles. |
| **Expected result** | Sequential lift cascade. Peak lift at T=750 ms when all four wings are in their respective power strokes simultaneously. Useful for vertical takeoff from a very confined stowed position. The aircraft can be stored fully stowed (wings alongside fuselage) and launch immediately with this sequence. |

UNIQUE TO THIS DESIGN

## PHASED DRAG BRAKING — SELECTIVE WAVE ARREST

| | |
|---|---|
| **Mechanisms used** | Rather than using all wings as airbrakes simultaneously (the quick-stop maneuver), wings are arrested sequentially from front to back — each wing contributes its drag pulse at a different moment. This smooths the braking force over time rather than delivering it as a single jolt, giving a controlled deceleration rather than an abrupt stop. The sequential braking can be spread over 2–4 seconds, giving a much smoother passenger experience at full scale. |

| | |
|---|---|
| **Wave parameters** | W1: drag AoA activated first. W2: drag AoA activated 250 ms later. W3: 500 ms later. W4: 750 ms later. Each wing: both strokes at 38–42 degrees simultaneously after activation. Duration: each wing holds drag mode for 1–2 seconds then returns to cruise. |
| **Expected result** | Smooth sustained deceleration rather than abrupt arrest. Deceleration rate: 3–5 m/s squared (versus 6–8 m/s squared for full arrest). Zero pitch attitude change — aircraft stays level throughout. At full scale this is the preferred landing approach mode — the smooth deceleration is comfortable for a passenger. |

UNIQUE TO THIS DESIGN

## INTER-WING VORTEX STEERING — WAKE SURFING

| | |
|---|---|
| **Mechanisms used** | The metachronal wave is deliberately timed so that each wing's power stroke begins exactly as the preceding wing's trailing edge vortex arrives at its leading edge. This is wake capture — the downstream wing surfs the upwash from the upstream wing's vortex, gaining lift for free. The timing is extremely precise: the phase offset is tuned to millisecond accuracy by the flight computer based on flight speed and wing spacing. It changes continuously with airspeed. |
| **Wave parameters** | Phase offsets: dynamically computed from airspeed and wing geometry. At hover (0 m/s): standard 90 degree spacing. At 10 km/h: phase spacing reduced to 75 degrees (vortex arrives sooner). At 30 km/h: phase spacing reduced to 55 degrees. Flight computer: updates phase table every 5 wing cycles. |
| **Expected result** | Estimated lift efficiency improvement of 8–15% versus fixed-phase operation. The improvement increases with forward speed. This is the closest thing to active aeroelastic wing coupling in a non-biological system. No other aircraft actively surfs the inter-wing vortex field with continuously-updated phase timing. |

UNIQUE TO THIS DESIGN

## CASCADE REVERSAL — MIRROR WAVE

| | |
|---|---|
| **Mechanisms used** | Left side and right side run in mirror-image waves — if the left side is at phase (0/90/180/270), the right side runs (270/180/90/0). The two sides are exact temporal mirrors of each other. The aerodynamic effect is that the lift pulses from left and right are precisely interleaved — when the left side has a peak lift moment, the right side is at its minimum, and vice versa. This produces a gentle rocking motion — a slow lateral oscillation at 1 Hz that can be used to deliberately redistribute load or to generate a lateral pumping motion for lateral translation. |
| **Wave parameters** | Left side: 0/90/180/270 degrees. Right side: 270/180/90/0 degrees (mirror). The lateral lift asymmetry alternates at 1 Hz. Net translation: can be biased left or right by amplitude asymmetry. |
| **Expected result** | 1 Hz lateral oscillation — gentle rocking in hover. With amplitude bias: slow lateral translation. The mirror wave is the most efficient pure lateral translation mode — it uses the natural timing asymmetry to generate lateral force rather than fighting the wave. Novel: no biological or engineering precedent for deliberate mirror-wave translation. |

# CHAPTER 14 — EXTENDED QUICK-REFERENCE TABLE

Wave parameters for all new maneuvers in this supplement. Refer to Volume 1, Chapter 9 for the original maneuver table.

| QUADCOPTER MANEUVERS | | | | | |
|---|---|---|---|---|---|
| **Maneuver** | **Phase** | **Amplitude** | **AoA** | **Sweep** | **Power** |
| Power loop | 0/90/180/270 | Symmetric | 35/15 inverted | Fwd stroke | High |
| Front flip | 0/90/180/270 | Symmetric | Inverted comp | Standard | High |
| Back flip | 0/90/180/270 | Symmetric | Inverted comp | Standard | High |
| Yaw spin | L leads R | Symmetric | Symmetric | Standard | Low |
| Inverted hover | 0/90/180/270 | Symmetric | 18-22 both | Standard | High |
| Matty flip | 0/90/180/270 | Symmetric | Nose-up arc | Varies | High |
| Split-S to hover | 0/90/180/270 | Half roll | Both-stroke | Varies | High |
| Punch out | 0/0/0/0 | Maximum | 42/18 both | Fwd max | High |
| Inverted yaw spin | L leads R | Symmetric | 18/22 inverted | Standard | High |
| Gap flying | 0/90/180/270 | Symmetric | Standard | W3W4 stow | Low |
| Orbit / inverted orbit | 0/90/180/270 | Symmetric | Std/inverted | Standard | Low-Med |
| **BIRD-INSPIRED** | | | | | |

| Maneuver | Phase | Amplitude | AoA | Sweep | Power |
|---|---|---|---|---|---|
| Pigeoning turn | Yaw-dominated | Symmetric | Both-stroke | Standard | Low |
| Thermal soaring | Stopped | Wings locked | 12 fixed | Fwd fixed | Zero |
| Slope soaring | Stopped | Wings locked | 12 fixed | Fwd fixed | Zero |
| Falcon stoop | All stowed | Stowed | Stowed | Full aft | Zero→High |
| Swift arrest | 0/180/0/180 | Symmetric | 40/40 drag | Aft tilt | High |
| Undulating flight | Alternating | Standard | Standard | Standard | Cyclic |
| Perch landing | 0/180/0/180 | Symmetric | Arrest→hover | Standard | Low |

| BAT AND INSECT-INSPIRED | | | | | |
|---|---|---|---|---|---|
| **Maneuver** | **Phase** | **Amplitude** | **AoA** | **Sweep** | **Power** |
| Dragonfly 180-turn | Max yaw diff | Symmetric | Standard | Standard | High |
| Figure-8 hover | 0/90/180/270 | Symmetric | Standard | Large dev | Low |
| Intercept dash | 0/90/180/270 | Asymmetric | Standard | Oblique | Med |
| Inertial flip | Stopped | L stow/R full | Stopped | Asymm | None |
| Inverted landing | 0/90/180/270 | Symmetric | Both-stroke | Standard | High |
| Sharp bat turn | 0/90/180/270 | Asymm+diff | Standard | Inside aft | Med |
| Hawkmoth station-keep | 0/180/0/180 | Symmetric | Standard | Standard | Low |

| | | | | | |
|---|---|---|---|---|---|
| Moth evasion | Random perturb | Random | Standard | Standard | Low |
| Escape jink | Single reverse | Maximum | Max | Standard | High |

| UNIQUE TO THIS DESIGN | | | | | |
|---|---|---|---|---|---|
| **Maneuver** | **Phase** | **Amplitude** | **AoA** | **Sweep** | **Power** |
| Accordion flight | 0→6→4→6→8 wings | Adapts | Standard | Prog stow | Med |
| Standing wave hover | 0/180/0/180 pairs | Reduced | 32/5 | Standard | Low |
| Asymmetric cascade | L=metachr R=ctr | Symmetric | Equal | Standard | Low |
| Sequential burst launch | Sequential deploy | Max at deploy | Max | 0→fwd | High |
| Phased drag braking | Sequential arrest | Symmetric | 38-42 both | Aft seq | High |
| Wake surfing | Speed-adaptive | Symmetric | Standard | Standard | Low |
| Mirror wave lateral | L=fwd R=mirror | Asymm bias | Standard | Standard | Low |

*KEY NOTE: Maneuvers marked with 'Unique to this design' have no direct equivalent in any currently operational aircraft or drone. They are made possible specifically by the combination of eight independently-controlled 3-DoF wings, the sweep-equals-stow mechanism, and the metachronal wave generator. Any single one of these elements in isolation would not enable these maneuvers — all three are required.*

*This supplement should be read together with Volume 1 of the Wave Function and Maneuver Control Guide. Together the two documents cover the complete maneuver envelope of the Metachronal Cascade Flapper from basic stable hover to maneuvers with no precedent in biology or engineering.*